Exam Date & Time: 23-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV/ DEC 2024
Saturday, 23rd November, 2024; 2.30 PM- 5.30 PM

MEDICAL DEVICES [BME 3122]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all the questions.

1A)	Identify the indirect method for the measurement of respiration. What is the advantage of this method over the direct method?	(4)
1B)	Illustrate with a figure, the principle and the instrumentation involved in Atomic absorption spectroscopy.	(3)
1C)	Explain the two conditioning units used in the oxygen therapy units. Also, indicate the purpose of each of these conditioning units in the oxygen therapy unit.	(3)
2A)	Propose how thoracic impedance can be used to monitor apnea events in newborns. Also, indicate the reason why alternating current is used as an excitation signal in these monitors.	(5)
2B)	Recommend the type of infrared detector suitable for maintaining constant sensitivity across a wide measurement range. Also, illustrate the working principle of the thermography equipment.	(3)
2C)	Find the pulse duration if a 2MHz transducer has 3 cycles in a pulse.	(2)
3A)	Illustrate with a neat figure, the working principle of Cathode Ray Tube used as a display unit in ultrasound equipment.	(4)
3B)	Define hematocrit and illustrate an automated method that uses laser light to count blood cells.	(4)
3C)	What happens to the power of the ultrasound beam, if the intensity of a sound beam remains unchanged, while the beam area is reduced to half?	(2)

4A)	Compare the different oxygenators used in the heart-lung machine.	(4)
4B)	Explain the principle of peritoneal dialysis. Also, mention its advantages and disadvantages.	(3)
4C)	What is ultrafiltration? How is it achieved in the dialysis unit? What are the factors that influence the degree of ultrafiltration?	(3)
5A)	Give reasons for the failure of the coil type and parallel plate dialyzer and explain with a neat figure, the most effective design of the hemodialyser.	(4)
5B)	With a neat flowchart, demonstrate the regulatory steps involved in selling or distributing medical devices.	(4)
5C)	What is the role of Central Licensing Authority (CLA) and State Licensing Authority (SLA) in medical device regulations?	(2)

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